Solutions for Week 8 Maths Exercise Sheet (Gaussian Elimination) + (Introduction to Vector Spaces)

Note that you will be given feedback for question (2).

(1) Solve the following system of four linear equations in four variables x_1, x_2, x_3, x_4 over the set of rational numbers:

$$x_1 + 5x_2 - 2x_3 + 3x_4 = -11$$

$$3x_1 - 2x_2 + 7x_3 + x_4 = 5$$

$$-2x_1 - x_2 - x_3 - 2x_4 = 0$$

$$5x_1 + 3x_2 + 4x_3 - x_4 = 13$$

Show all the intermediate steps (not just the final values obtained for x_1, x_2, x_3, x_4).
Comment: For practice, solve first using the "standard" elimination method and then also using the compact way (using a matrix). Once you obtain values for x_1, x_2, x_3, x_4 plug them back into each of the four equations to verify that your answer is correct.
Solution by "standard" elimination: First we rewrite the four given equations:

$$\begin{aligned} x_1 + 5x_2 - 2x_3 + 3x_4 &= -11 \quad (1)\\ 3x_1 - 2x_2 + 7x_3 + x_4 &= 5 \quad (2)\\ -2x_1 - x_2 - x_3 - 2x_4 &= 0 \quad (3)\\ 5x_1 + 3x_2 + 4x_3 - x_4 &= 13 \quad (4) \end{aligned}$$

We will now eliminate the variable x_1 to obtain three equations in the three variables x_2 and x_3.
Equation (4) minus five times equation (1) gives

$$-22x_2 + 14x_3 - 16x_4 = 68 \quad (5)$$

Two times equation (1) plus equation (3) gives

$$9x_2 - 5x_3 + 4x_4 = -22 \quad (6)$$

Two times equation (2) plus three times equation (3) gives

$$-7x_2 + 11x_3 - 4x_4 = 10 \quad (7)$$

Now we have three equations in three variables x_2, x_3, x_4. We will now eliminate x_4.
Equation (6) plus equation (7) gives

$$2x_2 + 6x_3 = -12 \quad (8)$$

Equation (5) plus four times equation (6) gives

$$14x_2 - 6x_3 = -20 \quad (9)$$

Adding equation (8) and equation (9) gives $16x_2 = -32$ which implies

$$x_2 = -2$$

Substituting this value of x_2 in equation (9) gives $-6x_3 = 8$ which implies

$$x_3 = \frac{-4}{3}$$

Substituting the values of x_2 and x_3 in equation (6) gives $-18-(\frac{-20}{3})+4x_4 = -22$ which implies

$$x_4 = \frac{-8}{3}$$

Substituting the values of x_2, x_3, x_4 in equation (1) gives $x_1+5(-2)-2(\frac{-4}{3})+3(\frac{-8}{3}) = -11$ which implies

$$x_1 = \frac{13}{3}$$

Solution using compact matrix form: We first write the four equations into a compact matrix form. Note that this matrix has four rows R_1, R_2, R_3, R_4 as read from top to bottom.

$$\left(\begin{array}{cccc|c} 1 & 5 & -2 & 3 & -11 \\ 3 & -2 & 7 & 1 & 5 \\ -2 & -1 & -1 & -2 & 0 \\ 5 & 3 & 4 & -1 & 13 \end{array}\right)$$

We are allowed only the following three operations (multiple times) on this matrix:

- Rearrange the rows
- Multiply (each entry) of a row by any rational number
- Add/subtract any row from another row

The aim is to convert the matrix into **row echelon form**. Using the row transformations given by $R_1 \to R_1, R_2 \to (R_2 - 3 \cdot R_1), R_3 \to (R_3 + 2 \cdot R_1), R_4 \to (R_4 - 5 \cdot R_1)$ we get

$$\left(\begin{array}{cccc|c} 1 & 5 & -2 & 3 & -11 \\ 0 & -17 & 13 & -8 & 38 \\ 0 & 9 & -5 & 4 & -22 \\ 0 & -22 & 14 & -16 & 68 \end{array}\right)$$

Using the row transformation given by $R_1 \to R_1, R_2 \to R_2, R_3 \to (9 \cdot R_2 + 17 \cdot R_3), R_4 \to (9 \cdot R_4 + 22 \cdot R_3)$ we get

$$\left(\begin{array}{cccc|c} 1 & 5 & -2 & 3 & -11 \\ 0 & -17 & 13 & -8 & 38 \\ 0 & 0 & 32 & -4 & -32 \\ 0 & 0 & 16 & -56 & 128 \end{array}\right)$$

Keeping the other rows as they are and doing $R_4 \to (R_3 - 2 \cdot R_4)$ gives

$$\left(\begin{array}{cccc|c} 1 & 5 & -2 & 3 & -11 \\ 0 & -17 & 13 & -8 & 38 \\ 0 & 0 & 32 & -4 & -32 \\ 0 & 0 & 0 & 108 & -288 \end{array}\right)$$

At this point, all entries of the matrix below the diagonal are zero. Hence, we can calculate the values of x_1, x_2, x_3, x_4 systematically as follows:

- First calculate value of x_4 using R_4
- Then using this value of x_4 calculate value of x_3 using R_3
- Then using the values of x_3, x_4 calculated earlier we can find value of x_2 using R_2
- Finally, using the values of x_2, x_3, x_4 calculated earlier we can find value of x_1 using R_1

However, we will further simplify the matrix to get it into **row echelon** form where the only non-zero entries on the left side of the matrix are on the diagonal.

Multiplying R_3 by $\frac{1}{32}$ and R_4 by $\frac{1}{108}$ we get

$$\left(\begin{array}{cccc|c} 1 & 5 & -2 & 3 & -11 \\ 0 & -17 & 13 & -8 & 38 \\ 0 & 0 & 1 & -1/8 & -1 \\ 0 & 0 & 0 & 1 & -8/3 \end{array}\right)$$

Keeping all other rows as they are and doing $R_3 \to (R_3 + \frac{1}{8} \cdot R_4)$ gives

$$\left(\begin{array}{cccc|c} 1 & 5 & -2 & 3 & -11 \\ 0 & -17 & 13 & -8 & 38 \\ 0 & 0 & 1 & 0 & -4/3 \\ 0 & 0 & 0 & 1 & -8/3 \end{array}\right)$$

Keeping all other rows as they are and doing $R_2 \to (R_2 - 13 \cdot R_3 + 8 \cdot R_4)$ gives

$$\left(\begin{array}{cccc|c} 1 & 5 & -2 & 3 & -11 \\ 0 & -17 & 0 & 0 & 34 \\ 0 & 0 & 1 & 0 & -4/3 \\ 0 & 0 & 0 & 1 & -8/3 \end{array}\right)$$

Keeping all other rows as they are and dividing R_2 by -17 gives

$$\left(\begin{array}{cccc|c} 1 & 5 & -2 & 3 & -11 \\ 0 & 1 & 0 & 0 & -2 \\ 0 & 0 & 1 & 0 & -4/3 \\ 0 & 0 & 0 & 1 & -8/3 \end{array}\right)$$

Keeping all other rows as they are and doing $R_1 \to (R_1 - 5 \cdot R_2 + 2 \cdot R_3 - 3 \cdot R_4)$ gives

$$\left(\begin{array}{cccc|c} 1 & 0 & 0 & 0 & 13/3 \\ 0 & 1 & 0 & 0 & -2 \\ 0 & 0 & 1 & 0 & -4/3 \\ 0 & 0 & 0 & 1 & -8/3 \end{array}\right)$$

From this matrix, we can now read out that $x_1 = 13/3, x_2 = -2, x_3 = -4/3$ and $x_4 = -8/3$.

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(2) (**feedback**) On Friday I bought two candy bars, a croissant and three coffees for 13.5 pounds. On Saturday, I bought three candy bars, five croissants and two coffees for 21.5 pounds. On Sunday (when I was very sleepy the whole day) I bought a candy bar, three croissants and six coffees for 26.5 pounds. Tomorrow, I wish to purchase four candy bars, two croissants and one coffee. How much will this cost me?

Comment: Show your working, i.e., don't just write down the final answer.
Solution: Let a, b, c be the cost of a candy bar, croissant and a coffee respectively. We get the following three equations from Friday, Saturday and Sunday:

$$\begin{aligned} 2a + b + 3c &= 13.5 \quad (10) \\ 3a + 5b + 2c &= 21.5 \quad (11) \\ a + 3b + 6c &= 26.5 \quad (12) \end{aligned}$$

We will now eliminate the variable a.

Two times equation (12) minus equation (10) gives

$$5b + 9c = 39.5 \quad (13)$$

Three times equation (12) minus equation (11) gives

$$4b + 16c = 58 \quad (14)$$

We now have two equations in two variables b and c. Using this, we will eliminate the variable b.

Four times equation (13) gives

$$20b + 36c = 158 \quad (15)$$

Five times equation (14) gives

$$20b + 80c = 290 \quad (16)$$

Equation (16) minus equation (15) gives $44c = 132 \Rightarrow c = 3$. Substituting value of c in equation (14) gives $4b = 10$ which implies $b = 2.5$. Substituting $b = 2.5$ and $c = 3$ in equation (12) gives $a = 1$.

Hence, the cost of a candy bar is $a = 1$ pound, the cost of a croissant is $b = 2.5$ pounds and the cost of a coffee is $c = 3$ pounds.

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(3) Let V be a vector space over a field F. Show each of the following:

(Q1) Additive identity $\vec{0}$ is unique

(Q2) $0\vec{v} = \vec{0}$

(Q3) $s\vec{0} = \vec{0}$

(Q4) $(-1)\vec{v} = \overrightarrow{-v}$

(Q5) $s\vec{v} = \vec{0}$ implies $s = 0$ or $\vec{v} = \vec{0}$

Remark 1: This is a bit harder question than the others in this exercise sheet. Some of the proofs are bit long to write, but the ideas are not too difficult/creative.
Remark 2: The addition and multiplication of elements within F is denoted by $+$ and $\times$ respectively. The addition of two vectors from V is via the addition operation denoted by $\oplus$. The multiplication of a scalar s and a vector $\vec{v}$ is denoted by $s\vec{v}$.
Comment: To prove the above, you can use the following:

- Any of the 8 conditions available due to V being a vector space over F
- Any of the conditions available due to F being a field

Solution: Let $(+, \times)$ denote the addition and multiplication in the field F. Let $\oplus$ denote addition of elements from V. Since V is a vector space over the field F, for any vectors $\vec{u}, \vec{v}, \vec{w} \in V$ and any scalars $r, s \in F$ we have the following 8 **conditions**:

(1) Commutativity of vector addition: $\vec{u} \oplus \vec{v} = \vec{v} \oplus \vec{u}$
(2) Associativity of vector addition: $(\vec{u} \oplus \vec{v}) \oplus \vec{w} = \vec{u} \oplus (\vec{v} \oplus \vec{w})$
(3) Existence of Additive identity: $\vec{0} \oplus \vec{v} = \vec{v}$
(4) Existence of additive inverse: for each $\vec{x}$, there exists $\overrightarrow{-x}$ such that $\vec{x} \oplus \overrightarrow{-x} = \vec{0}$
(5) Associativity of multiplication of scalar & vector: $r(s\vec{v}) = (rs)\vec{v}$
(6) Distributivity of scalar sums: $(r+s)\vec{v} = r\vec{v} \oplus s\vec{v}$
(7) Distributivity of vector sums: $r(\vec{u} \oplus \vec{v}) = r\vec{u} \oplus r\vec{v}$
(8) Existence of identity of multiplication of scalar & vector: $1\vec{v} = \vec{v}$

where vector addition and multiplication of a scalar with a vector are defined as follows:

- Vector addition: for each $\vec{u}, \vec{v}$ a vector from V is assigned to $\vec{u} \oplus \vec{v}$
- Multiplication of a scalar by a vector: for each $s \in F$ and $\vec{v} \in V$, a vector from V is assigned to $s\vec{v}$

We are now ready to prove the five equations: (can you come up with simpler/shorter proofs?)

(Q1) Additive identity $\vec{0}$ is unique: Suppose there is another additive identity $\vec{a} \in V$, i.e.,

$$\vec{a} \oplus \vec{v} = \vec{v} \text{ for all } \vec{v} \in V \tag{17}$$

Substituting $\vec{v} = \vec{0}$ in equation (17) gives $\vec{a} \oplus \vec{0} = \vec{0}$. By Condition (3) it follows that $\vec{a} \oplus \vec{0} = \vec{a}$. Hence, $\vec{a} = \vec{0}$.

(Q2) $0\vec{v} = \vec{0}$: Let $\vec{v} \in V$ be any vector. We obtain the following series of equalities:

$$\begin{aligned}
\vec{0} &= \vec{v} \oplus \overrightarrow{-v} && \text{existence of } \overrightarrow{-v} \text{ is given by Condition (4)} \\
&= \overrightarrow{-v} \oplus \vec{v} && \text{by Condition (1)} \\
&= \overrightarrow{-v} \oplus 1\vec{v} && \text{by Condition (8)} \\
&= \overrightarrow{-v} \oplus (1+0)\vec{v} && \text{by neutral element for adddition in } F \\
&= \overrightarrow{-v} \oplus \Big(1\vec{v} \oplus 0\vec{v}\Big) && \text{by Condition (6)} \\
&= \overrightarrow{-v} \oplus \Big(\vec{v} \oplus 0\vec{v}\Big) && \text{by Condition (8)} \\
&= \Big(\overrightarrow{-v} \oplus \vec{v}\Big) \oplus 0\vec{v} && \text{by Condition (2)} \\
&= \vec{0} \oplus 0\vec{v} && \text{by Condition (4)} \\
&= 0\vec{v} && \text{by Condition (3)}
\end{aligned}$$

(Q3) $s\vec{0} = \vec{0}$: By Condition (4), there exists $\overrightarrow{-0} \in V$ such that $\vec{0} \oplus \overrightarrow{-0} = \vec{0}$. By Condition (3), $\vec{0} \oplus \overrightarrow{-0} = \overrightarrow{-0}$. Therefore, it follows that $\overrightarrow{-0} = \vec{0}$.
Let $s \in F$ be any scalar. Then we have

$$\begin{aligned}
s\vec{0} &= s(\vec{0} \oplus \overrightarrow{-0}) && \text{since } \vec{0} \oplus \overrightarrow{-0} = \vec{0} \\
&= s\vec{0} \oplus s\overrightarrow{-0} && \text{by Condition (7)} \\
&= s\vec{0} \oplus s\vec{0} && \text{since } \vec{0} = \overrightarrow{-0}
\end{aligned}$$

Therefore, we have

$$s\vec{0} = s\vec{0} \oplus s\vec{0} \tag{18}$$

Let $s\vec{0} = \vec{w}$. By Condition (4), there exists $\overrightarrow{-w} \in V$ such that $\vec{w} \oplus \overrightarrow{-w} = \vec{0}$. Adding $\overrightarrow{-w}$ to both sides of equation (18), we get $\vec{w} \oplus \overrightarrow{-w} = (\vec{w} \oplus \vec{w}) \oplus \overrightarrow{-w}$. By Condition (4), left-hand side of this equation is equal to $\vec{0}$. By Condition (2), the right-hand side of this equation is equal to $\vec{w} \oplus (\vec{w} \oplus \overrightarrow{-w}) = \vec{w} + \vec{0} = \vec{w}$ where we have used Condition (4) and then Condition (3) in the last two equalities. Hence, $s\vec{0} = \vec{w} = \vec{0}$.

(Q4) $\underline{(-1)\vec{v} = \overrightarrow{-v}}$: We obtain the following series of equalities:

$$\begin{aligned}
\overrightarrow{-v} &= \vec{0} \oplus \overrightarrow{-v} && \text{by Condition (3)} \\
&= \overrightarrow{-v} \oplus \vec{0} && \text{by Condition (1)} \\
&= \overrightarrow{-v} \oplus 0\vec{v} && \text{since we have shown } 0\vec{v} = \vec{0} \text{ in second part (Q2) of this question} \\
&= \overrightarrow{-v} \oplus \Big(1 + (-1)\Big)\vec{v} && \text{by existence of additive inverse of 1 in } F \\
&= \overrightarrow{-v} \oplus \Big(1\vec{v} \oplus (-1)\vec{v}\Big) && \text{by Condition (6)} \\
&= \overrightarrow{-v} \oplus \Big(\vec{v} \oplus (-1)\vec{v}\Big) && \text{by Condition (8)} \\
&= \Big(\overrightarrow{-v} \oplus \vec{v}\Big) \oplus (-1)\vec{v} && \text{by Condition (2)} \\
&= \vec{0} \oplus (-1)\vec{v} && \text{by Condition (4)} \\
&= (-1)\vec{v} && \text{by Condition (3)}
\end{aligned}$$

(Q5) $\underline{s\vec{v} = \vec{0} \text{ implies } s = 0 \text{ or } \vec{v} = \vec{0}}$: If $s = 0$ we are done. So suppose $s \neq 0$. Then by existence of multiplicative inverse in F, there exists $s^{-1} \in F$ such that $s^{-1}s = 1$ where 1 is multiplicative identity in F. We now show that $\vec{v} = \vec{0}$

$$\begin{aligned}
\vec{v} &= 1\vec{v} && \text{by Condition (8)} \\
&= (s^{-1}s)\vec{v} && \text{since } s^{-1}s = 1 \\
&= s^{-1}(s\vec{v}) && \text{by Condition (5)} \\
&= s^{-1}\vec{0} && \text{since } s\vec{v} = \vec{0} \\
&= \vec{0} && \text{by the third part (Q3) of this question}
\end{aligned}$$

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(4) Verify that $\mathbb{Q}^2$ is a vector space over the field $\mathbb{Q}$ with the following definition of vector addition and scalar multiplication:

(a) Vector addition: If $\vec{v_1} = \binom{a_1}{b_1}$ and $\vec{v_2} = \binom{a_2}{b_2}$ then $\vec{v_1} \oplus \vec{v_2}$ is defined to be $\binom{a_1+a_2}{b_1+b_2}$

(b) Multiplication of a scalar with a vector: $\vec{v} = \binom{a}{b} \in \mathbb{Q}^2$ and $r \in \mathbb{Q}$ then $r\vec{v} = \binom{r \times a}{r \times b}$

Comment: Each of the 8 conditions in the definition of vector space over a field needs to be verified. You can use any of the conditions available due to $\mathbb{Q}$ being a field.

Remark : The addition and multiplication of elements within $\mathbb{Q}$ is denoted by $+$ and $\times$ respectively. The addition of two vectors from $\mathbb{Q}^2$ is via the addition operation denoted by $\oplus$. The multiplication of a scalar s and a vector $\vec{v}$ is denoted by $s\vec{v}$.

Solution:

Recall that Q being a field gives us the following properties we can use for solving this question:

$$
\begin{array}{rclrcll}
a+0 &=& a & a\times 1 &=& a & \text{(neutral elements)}\\
a+b &=& b+a & a\times b &=& b\times a & \text{(commutativity)}\\
(a+b)+c &=& a+(b+c) & (a\times b)\times c &=& a\times(b\times c) & \text{(associativity)}\\
a+(-a) &=& 0 & a\times a^{-1} &=& 1\ \ (a\neq 0) & \text{(inverses)}
\end{array}
$$

$$a\times(b+c) = a\times b + a\times c \qquad \text{(distributivity of } \times \text{ over } +)$$

Having defined vector addition & multiplication of a scalar and a vector, we are now ready to verify the 8 conditions for $\mathbb{Q}^2$ to be a vector space over the field $\mathbb{Q}$ of rational numbers. Let $\vec{u} = \binom{a}{b}, \vec{v} = \binom{c}{d}$ and $\vec{w} = \binom{e}{f}$ for some $a, b, c, d, e, f \in \mathbb{Q}$. Let $r, s \in \mathbb{Q}$ be scalars.

(1) <u>Commutativity of vector addition:</u>
$\vec{u} \oplus \vec{v} = \binom{a}{b} \oplus \binom{c}{d} = \binom{a+c}{b+d} = \binom{c+a}{d+b} = \binom{c}{d} \oplus \binom{a}{b} = \vec{v} \oplus \vec{u}$
where we have used commutativity of addition in the field $\mathbb{Q}$ to write $a+c = c+a$ and $b+d = d+b$.

(2) <u>Associativity of vector addition</u>:

$$(\vec{u} \oplus \vec{v}) \oplus \vec{w} = \left(\begin{pmatrix} a \\ b \end{pmatrix} \oplus \begin{pmatrix} c \\ d \end{pmatrix}\right) \oplus \begin{pmatrix} e \\ f \end{pmatrix} = \begin{pmatrix} a+c \\ b+d \end{pmatrix} \oplus \begin{pmatrix} e \\ f \end{pmatrix} = \begin{pmatrix} (a+c)+e \\ (b+d)+f \end{pmatrix}$$

$$= \begin{pmatrix} (a+(c+e) \\ (b+(d+f) \end{pmatrix} = \begin{pmatrix} a \\ b \end{pmatrix} \oplus \begin{pmatrix} c+e \\ d+f \end{pmatrix} = \begin{pmatrix} a \\ b \end{pmatrix} \oplus \left(\begin{pmatrix} c \\ d \end{pmatrix} \oplus \begin{pmatrix} e \\ f \end{pmatrix}\right) = \vec{u} \oplus (\vec{v} \oplus \vec{w})$$

(3) <u>Existence of Additive identity</u>: We claim that $\binom{0}{0} \in \mathbb{Q}^2$ is an additive identity. Consider any $\vec{x} = \binom{x_1}{x_2} \in \mathbb{Q}^2$. Then we have
$\vec{0} \oplus \vec{x} = \binom{0}{0} \oplus \binom{x_1}{x_2} = \binom{0+x_1}{0+x_2} = \binom{x_1}{x_2} = \vec{x}$ where we have used the fact that 0 is an additive identity (i.e., neutral element) for addition in the field $\mathbb{Q}$.

(4) <u>Existence of additive inverse</u>: Consider any $\vec{x} = \binom{x_1}{x_2} \in \mathbb{Q}^2$. We claim that the vector $\binom{-x_1}{-x_2}$ is an additive inverse for x where $-x_1, -x_2 \in \mathbb{Q}$ are the additive inverses of x_1, x_2 respectively in the field $\mathbb{Q}$. Hence, we have

$$\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \oplus \begin{pmatrix} -x_1 \\ -x_2 \end{pmatrix} = \begin{pmatrix} x_1 + (-x_1) \\ x_2 + (-x_2) \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} = \vec{0}$$

where we have used the fact that $x_1 + (-x_1) = 0 = x_2 + (-x_2)$ since $-x_1, -x_2 \in \mathbb{Q}$ are the additive inverses of x_1, x_2 respectively in the field $\mathbb{Q}$.

(5) <u>Associativity of multiplication of scalar & vector</u>: Let $\vec{v} = \binom{c}{d}$ be any vector from $\mathbb{Q}^2$. Then we have

$$r(s\vec{v}) = r(s\begin{pmatrix} c \\ d \end{pmatrix}) = r(\begin{pmatrix} s\times c \\ s\times d \end{pmatrix}) = \begin{pmatrix} r\times(s\times c) \\ r\times(s\times d) \end{pmatrix} = \begin{pmatrix} (r\times s)\times c \\ (r\times s)\times c \end{pmatrix} = (r\times s)\begin{pmatrix} c \\ d \end{pmatrix} = (rs)\vec{v}$$

where we have used the associativity of the multiplication operation $\times$ in the field $\mathbb{Q}$.

(6) Distributivity of scalar sums: Let $\vec{v} = \binom{a}{b}$ be any vector in $\mathbb{Q}^2$. Then we have

$$(r+s)\vec{v} = (r+s)\begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} (r+s) \times a \\ (r+s) \times b \end{pmatrix} = \begin{pmatrix} r \times a + s \times a \\ r \times b + s \times b \end{pmatrix} = \begin{pmatrix} r \times a \\ r \times b \end{pmatrix} \oplus \begin{pmatrix} s \times a \\ s \times b \end{pmatrix}$$
$$= r\begin{pmatrix} a \\ b \end{pmatrix} \oplus s\begin{pmatrix} a \\ b \end{pmatrix} = r\vec{v} \oplus s\vec{v}$$

where we have used distributivity of the multiplication operation $\times$ over the addition operation $+$ in the field $\mathbb{Q}$

(7) Distributivity of vector sums: Let $\vec{u} = \binom{a}{b}, \vec{v} = \binom{c}{d}$ and $\vec{w} = \binom{e}{f}$ be any two vectors from $\mathbb{Q}^2$. Then we have

$$r(\vec{u} \oplus \vec{v}) = r\Big(\begin{pmatrix} a \\ b \end{pmatrix} \oplus \begin{pmatrix} c \\ d \end{pmatrix}\Big) = r\Big(\begin{pmatrix} a+c \\ b+d \end{pmatrix}\Big) = \begin{pmatrix} r \times (a+c) \\ r \times (b+d) \end{pmatrix} = \begin{pmatrix} (r \times a) + (r \times c) \\ (r \times b) + (r \times d) \end{pmatrix}$$
$$= \begin{pmatrix} r \times a \\ r \times b \end{pmatrix} \oplus \begin{pmatrix} r \times c \\ r \times d \end{pmatrix} = r\vec{u} \oplus r\vec{v}$$

where we have used distributivity of the multiplication operation $\times$ over the addition operation $+$ in the field $\mathbb{Q}$

(8) Existence of identity of multiplication of scalar & vector: $1\vec{v} = \vec{v}$
Since Q is a field we have a multiplicative identity (i.e., neutral element) 1 for multiplication of two scalars from Q. That is, $1 \times q = q$ for each $q \in \mathbb{Q}$. We now claim that this same 1 is also a multiplicative identity for multiplication of a scalar and a vector. Let $\vec{v} = \binom{a}{b}$ be any vector in $\mathbb{Q}^2$. Then we have

$$1\vec{v} = 1\begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} 1 \times a \\ 1 \times b \end{pmatrix} = \begin{pmatrix} a \\ b \end{pmatrix} = \vec{v}$$

where we have used the fact that $1 \times a = a$ and $1 \times b = b$.